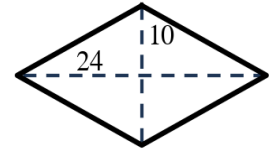


Speak Like A Mathematician:

apothem:

central angle of a regular polygon:



Area Formulas for Polygons

Rhombus or kite: $A = \frac{1}{2} d_1 \text{ _____}$ (half the product of the diagonals).

Regular polygon: $A = \frac{1}{2} a \text{ _____}$, where a is the apothem and P is the perimeter.

To find the apothem, use the right triangle: half the central angle sits at the center, and $\tan(\text{half-angle}) = (\text{half-side}) / \text{_____}$.

Example 1 - Rhombus

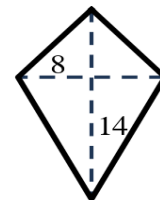
The rhombus at the top right has diagonals 10 and 24.

a) Find its area.

Example 2 - Kite

The kite at the right has diagonals 8 and 14.

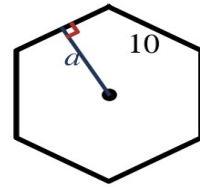
a) Find its area.



Example 3 - Regular Hexagon (Exact)

The hexagon at the right has sides of length 10.

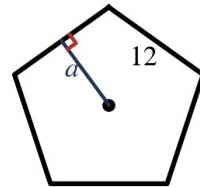
- Find the apothem exactly. (Hint: a hexagon's central triangles are equilateral — the apothem is a 30-60-90 long leg.)
- Find the area exactly, then to the nearest tenth.



Example 4 - Regular Pentagon (Trig)

The pentagon at the right has sides of length 12.

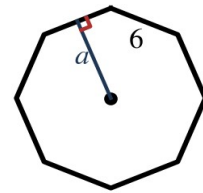
- Find the central angle, and the half-angle at the top of one central triangle.
- Find the apothem, then the area (nearest tenth).



Example 5 - Regular Octagon

The octagon at the right has sides of length 6.

- Find the half central angle.
- Find the apothem and the area (nearest tenth).



Example 6 - Central Angles

- Find the central angle of a regular nonagon (9 sides).
- Why does the apothem method work for ANY regular polygon?